AIRLINE RESERVATION SYSTEM

A Real-Time Research Project Report Submitted to
JNTU Hyderabad in Partial Fulfillment of the
Requirement for the award of the Degree

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MALLA REDDY INSTITUTE OF TECHNOLOGY AND SCIENCE
Approved by AICTE New Delhi and Affiliated to JNTU
An ISO 9001:2015 Certified Institution
Accredited by NBA and NAAC 'A'Grade
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CERTIFICATE

This is to certify that the Real-Time Research Project entitled **"AIRLINE RESERVATION SYSTEM"** has been submitted by **MOTIKE SREESHAILAM (22S11A05JO), TEKULA NAVEENKUMAR (22S11A05F7), THALLA VIGNESH (22S11A05J7)** in partial fulfilment of the requirements for the award of **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE AND ENGINEERING**. **This record of bonafide work was carried out by them under my guidance and supervision.** ***The result embodied in this Real-Time Research project report has not been submitted to any other University or Institute for the award of any degree.***

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The Real-Time Research Project work carried out by our team in the Department of Computer Science and Engineering, Malla Reddy Institute of Technology and Science, Hyderabad. ***This work is original and has not been submitted in part or full for any degree or diploma of any other university.***

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ABSTRACT

The main purpose of this application is to reduce the manual errors involved in the airline reservation process. And make it environment for the customers to book the flights as when they require such they can utilize this application to make reservation, modify reservations or cancel a particular reservation.

An airline reservation system is a sophisticated software solution designed to facilitate the booking and management of airline tickets. It enables customers to search for flights, check availability, compare prices, and make reservations online. The system integrates with airline schedules, fare information, and payment gateways to provide real-time data and secure transactions. Key functionalities include managing passenger information, issuing electronic tickets, handling cancellations, and making itinerary changes. By automating these processes, the airline reservation system enhances operational efficiency for airlines and offers a seamless, user-friendly experience for travelers, improving customer satisfaction and streamlining the overall travel process. Airline reservations system is an integrated passenger processing system, including inventory, fares, ticket-less operations and credit card transactions. All communications are via TCP/IP network protocol enabling the using of both intranet and internet communications worldwide. The solution includes several standard items, which are combined to provide an integrated solution with interfaces to other business systems. The system is based on open architecture, using industry standard equipment and software.Airline reservations system is an integrated passenger processing system, including inventory, fares, ticket-less operations and credit card transactions. All communications are via TCP/IP network protocol enabling the using of both intranet and internet communications worldwide.The solution includes several standard items, which are combined to provide an integrated solution with interfaces to other business systems. The system is based on open architecture, using industry standard equipment and software.

CHAPTER-1
INTRODUCTION

Imagine a world where booking a flight involved countless phone calls and visits to travel agents, each with limited access to real-time information. This was the reality before airline reservation systems (ARS) revolutionized air travel in the late 1950s. These early systems were a vast improvement, centralizing flight schedules, fares, and seat availability for internal airline use. Airline staff could efficiently manage reservations and ensure accurate passenger information.The industry boomed further with the rise of computer reservation systems (CRS) essentially more sophisticated versions of the ARS. CRSs not only streamlined internal airline operations but also opened the door for wider distribution channels. Travel agencies gained access to a vast network of airlines through a single platform, allowing them to efficiently book flights for their clients. This innovation significantly improved the customer experience, making it easier than ever to compare fares, find flights, and secure reservations.

1.1 Problem Definition&Description

Airline reservation systems often face challenges in managing complex data (flight schedules, fares, passenger information) and providing a smooth user experience. The problem lies in creating a system that is efficient, accurate, and user-friendly. An ideal system should streamline the reservation process for customers (booking flights, managing reservations, online payments) while also offering real-time information and inventory control for the airline. This requires a robust system that integrates various functionalities seamlessly to ensure a positive experience for both passengers and airline staff seamlessly to ensure a positive experience for both passengers and airline staff.

1.2 Objectives of Project

processes An airline reservation system project aims to develop a software solution that streamlines flight booking for both airlines and passengers. The core objectives typically focus on creating a user-friendly platform for travelers to search flights, compare fares, and book tickets efficiently. Additionally, the system should effectively manage flight inventory, passenger information, and reservation data for the airline, improving operational efficiency. Overall, the project strives to enhance the travel booking experience for customers while providing airlines with a robust system for managing reservations and flight information.

1.3 Scope of the Project

The airline reservation system project aims to develop a comprehensive software application that streamlines air travel booking and management, catering to both passengers and airline personnel. Passengers will be able to search for flights based on origin, destination, and date, view available fares and schedules, select seats, book tickets, and manage their reservations.

CHAPTER-2
System Analysis

2.1 Existing System

In the existing system data is entered manually. When records are accessed frequently managing such records difficult. Therefore organizing data becomes difficult.These systems are designed to manage real-time flight information, seat availability, pricing, and passenger data. Major airlines often use global distribution systems (GDS) such as Amadeus, Sabre, and Travelport, which connect to various airlines, travel agencies, and other service. providers. This allows customers to search for flights, compare options, book tickets, and make payments seamlessly online.

2.1.1 Advatanges of Existing System

i) Travelers, the system offers unparalleled convenience and choice.

ii) They can search and compare flights across multiple airlines quickly and easily, finding the best option for their needs.

iii) Additionally, features like online booking and seat selection allow for a smoother travel experience.

2.1.2 Disadvatanges of Existing System

i)Modifications are complicated.

ii)Much time consuming.

iii) complexity of the system.

iv) Travelers often struggle with limited search options when usoinsgt GDS platforms, making it difficult to find the suitable fares and combinations for multi-city trips or complex itineraries.

2.2 Proposed System

The proposed system is better and more efficient than existing system by overcome all the drawbacks of the present system to permanent solution to them. The primary aim of the new system is to speed up the transactions. User friendliness is another peculiarity of the proposed system. The main advantage of the proposed system is the reduction in labour as it will be possible to search the details of the various flights information .Another important feature of the proposed system is the data security is provided by the system.

2.2.1 Advatanges of Proposed System

- Complex functions are done automatically.
- Processing time can be minimized
- Simple and easy to manage
- Changes of errors reduced
- Faster and more accurate
- We have more concession in ticket fair for senior citizen

2.3 Software and Hardware requirements

2.3.1 Software Requirements

1)Flight Management:

i) Ability to define flight schedules, routes, fares, and seat inventory. ii)Real-time updates on flight availability, delays, and

2)Reservation Management:

i) User-friendly interface for searching flights based on origin, destination, date, and passenger details.

ii) Secure booking process with options for seat selection, special requests (meals, disabilities), and fare types.

3) Customer Management:

i) Secure user accounts for storing traveler information and preferences.

ii)Tracking of frequent flyer miles and loyalty programs.

4) _Reporting and Analytics:

i) Generation of reports on flight bookings, revenue, passenger demographics, and popular routes.

ii) Data analysis tools to understand travel trends and optimize pricing strategies.

5) Performance:

i) Fast response times for searches, bookings, and other actions. ii)Ability to handle high volumes of traffic during peak booking periods.

2.3.2 Hardware Requirements

Servers:

i) The core of the system resides on servers.

ii) These machines handle data processing, user requests, and communication with other systems. **iii)** High-performance servers with multiple processors and ample RAM are crucial for handling peak booking times and ensuring smooth operation. **Storage:**

i) A robust storage solution is needed to hold the airline's flight data, customer information, and booking history.

ii) This could involve a combination of hard disk drives (HDDs) for bulk storage and solid- state drives (SSDs) for faster access to frequently used data.

Networking:

i) Airline reservation systems rely heavily on reliable networking to connect various components and enable communication with external systems like GDS.

ii) High-bandwidth connections and network redundancy are essential for ensuring system uptime and responsiveness.

Backup and Disaster Recovery:

i) Safeguarding critical data is paramount.

ii) Backup systems and a disaster recovery plan are necessary to recover from hardware failures or unforeseen events.

2.4 System Configuration:-

2.4.1. Hardware System Configuration:-

Processor	- Pentium –III
Speed	- 1.1 Ghz
RA	- 256 MB(min)
M Hard	- 20 GB - 1.44 MB - Standard
Disk	Windows Keyboard - Two or
Floppy Drive	Three Button Mouse
Key Board	
Mouse	- SVGA
Monitor	

2.4.2. Software System Configuration:-

Operating System	:
Application Server	Windows95/98/2000/XP
Front End	: Tomcat5.0/6.X
	: HTML, Java, Jsp
Scripts	: JavaScript.
Server side Script	: Java Server Pages.
Database	: Mysql

Database Connectivity : JDBC.

2.5 System Study:-

2.5.1 Feasibility Study:-

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

2.5.2 Economical Feasibility :-

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

2.5.3 Technical Feasibility :-

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

2.5.4 Social Feasibility :-

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER -3

Architectural Design

3.1 Modules Design :-

An airline reservation system can be broken down into several key modules to handle different functionalities. Here's a breakdown of some common modules:

3.1.1 User Management:

Registration: Allows new users to create accounts for online booking.
Login: Authenticates existing users for secure access to their profiles and bookings. **Profile Management:** Enables users to update personal information, preferences, and frequent flyer details
.

3.1.2 Flight Management:

Flight Search: Provides functionalities to search for flights by origin, destination, date, and other criteria. This might integrate with a Global Distribution System (GDS) for real- time information. **Flight Schedule:** Displays current and upcoming flight schedules with details like departure/arrival times, duration, and layover information. **Inventory Management:** Tracks available seats for each flight class (economy, business, etc.) and ensures real-time updates to prevent overbooking.

3.1.3 Reservation Management:

Seat Selection: Allows users to choose their preferred seats during the booking process. **Fare Calculation:** Calculates the total fare for the chosen flight based on factors like class, baggage allowance, and any applicable taxes or fees.
Payment Processing: Integrates with secure payment gateways to handle online transactions for booking confirmation.

3.1.4 Additional Modules:

Customer Relationship Management (CRM): Tracks customer interactions, purchase history, and preferences to personalize future offerings and promotions.
Reporting and Analytics: Generates reports on booking trends, revenue, passenger demographics, and other valuable data for informed decision-making.

Administrative Panel: Provides functionalities for airline staff to manage flight schedules, fares, promotions, and access user data for maintenance purposes.

Remember, this is a general structure, and specific functionalities might vary depending on the airline's needs.

3.2 INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

What data should be given as input?

How the data should be arranged or coded?

The dialog to guide the operating personnel in providing input. Methods for preparing input validations and steps to follow when error occur.

3.3 OBJECTIVES

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.
2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free fdraotma errors. The data entry screen is designed in such a way that all manipulates the can be performed. It also provides record viewing facilities.
3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needded so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow

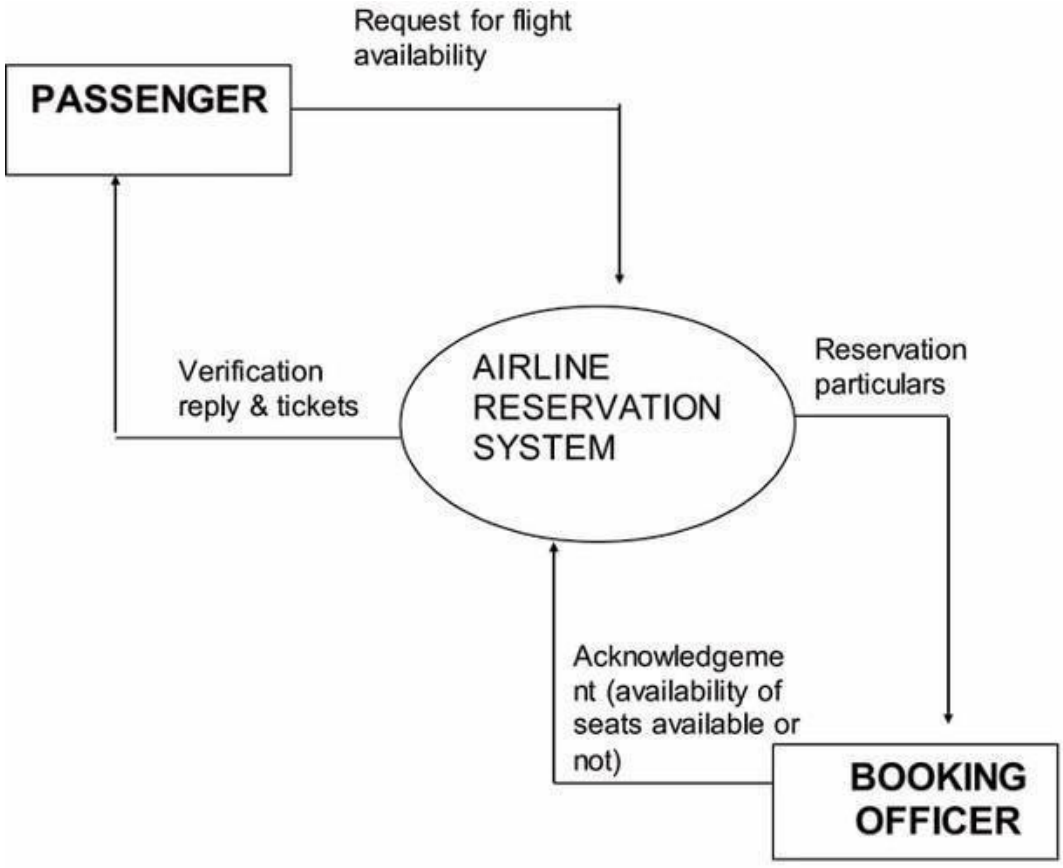
3.4 OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information

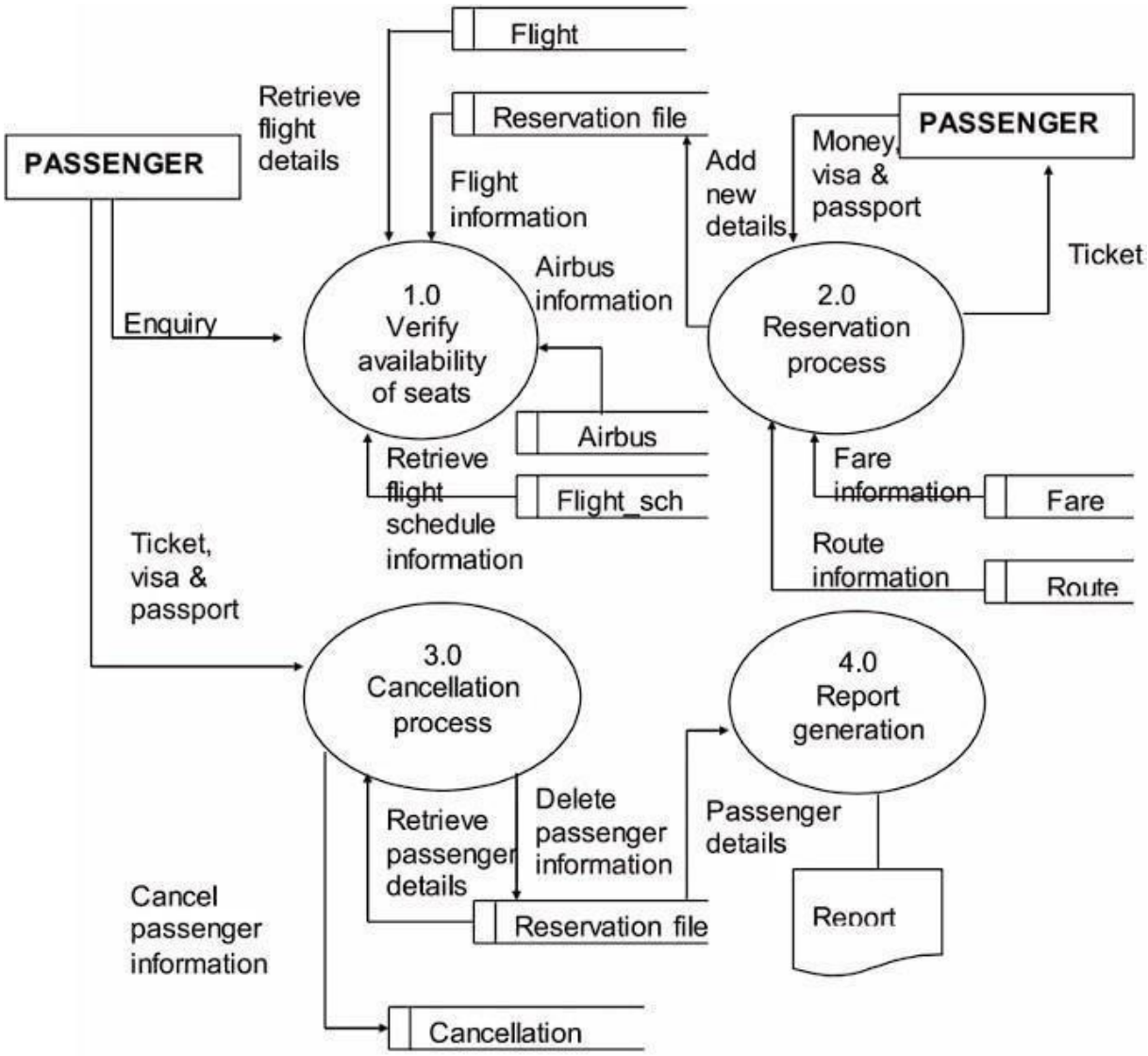
to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making. 1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements. 2.Select methods for presenting information. 3.Create document, report, or other formats that contain information produced by the system. The output form of an information system should accomplish one or more of the following objectives.

Convey information about past activities, current status or projections of the Future. Signal important events, opportunities, problems, or warnings. Trigger an action. Confirm an action.

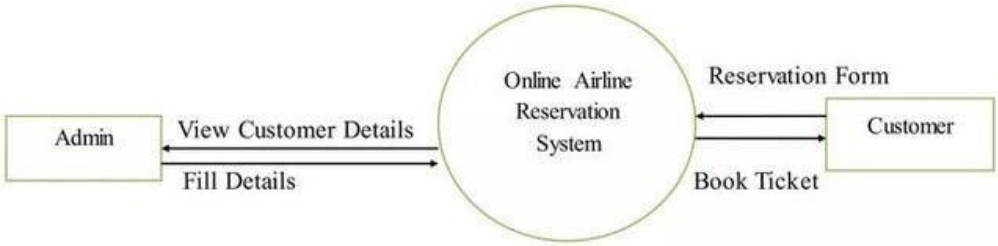
3.5 context diagram for Airline Reservation System



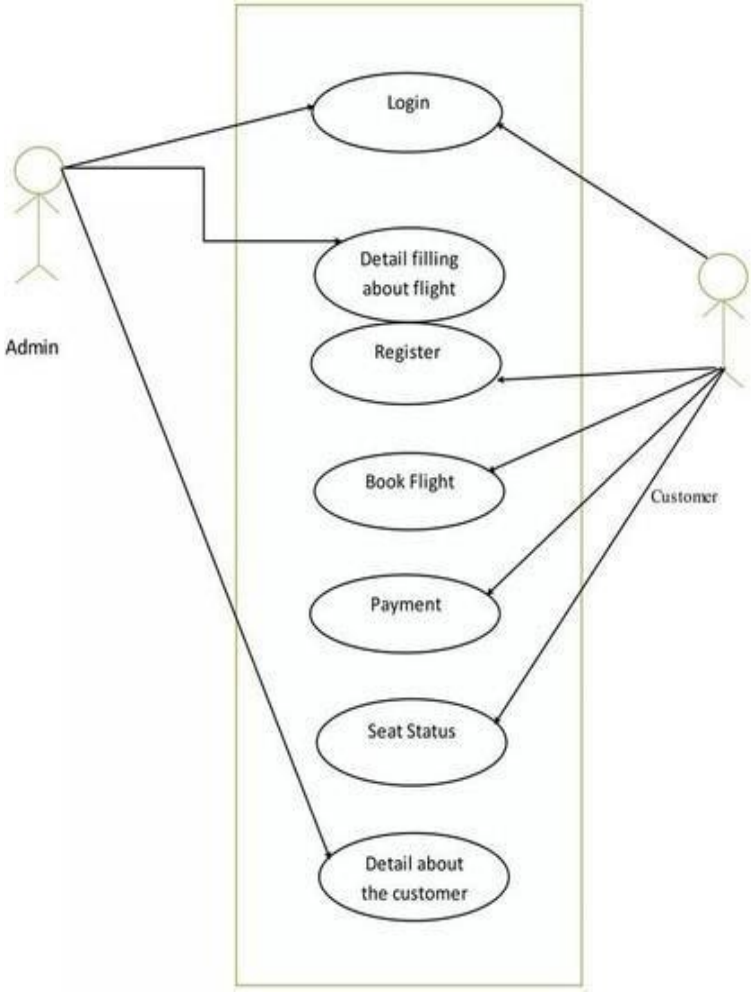
3.6 Data Flow Diagram



3.7. Context Diagram



3.8 Use Case Diagram



CHAPTER -4
IMPLEMENTING &TESTING

4.1 Software Environment

4.1.1 Java Technology

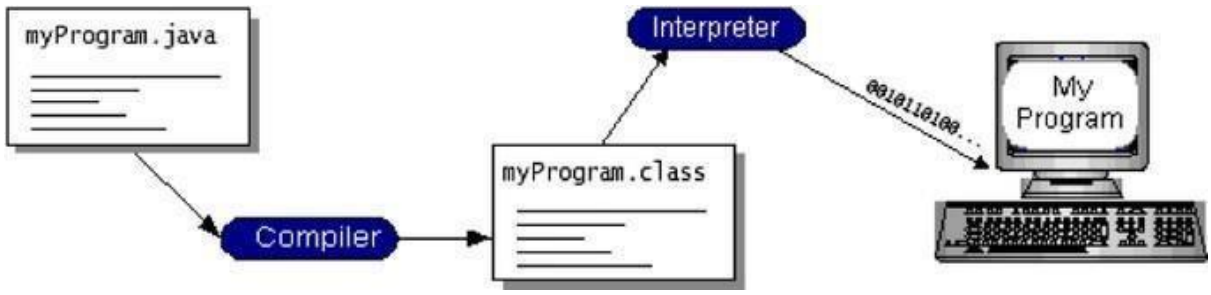
Java technology is both a programming language and a platform. The

Java Programming Language
The Java programming language is a high-level language that can be characterized by all of the following buzzwords:

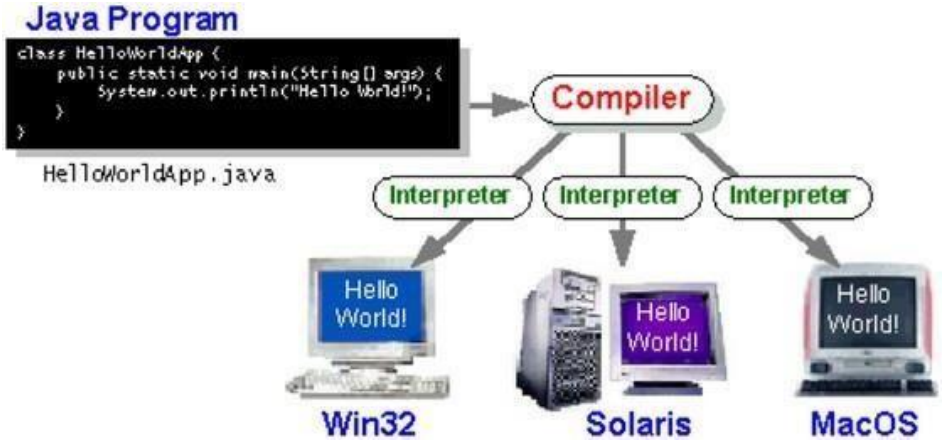
Simple
Architecture neutral
Object oriented Portable
Distributed
High performance
Interpreted
Multithreaded
Robust
Dynamic Secure

With most programming languages, you either compile or interpret a program so that you can run it on your computer. The Java programming language is unusual in that a
program is both compiled and interpreted. With the compiler, first you translate a program into an intermediate language called Java byte codes —the platform-independent codes interpreted by the interpreter on the Java platform. The interpreter parses and runs each Java byte code instruction on the computer.

4.1.2. Compilation happens just once; Interpetation occurs each time the program is executed. The following figure illustrates how this works.



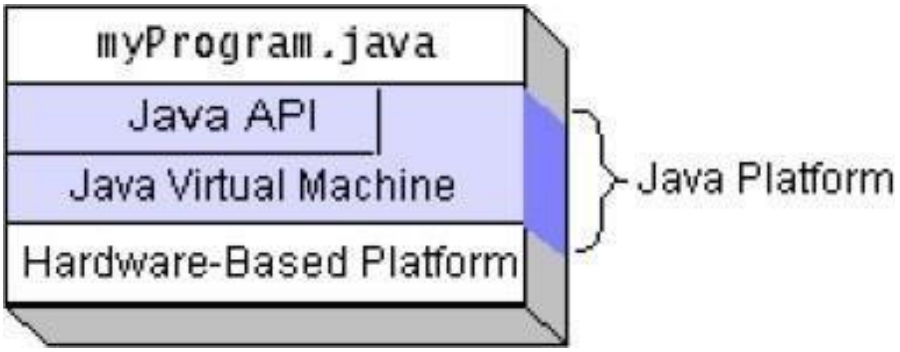
You can think of Java byte codes as the machine code instructions for the Java Virtual Machine (Java VM). Every Java interpreter, whether it's a development tool or a Web browser that can run applets, is an implementation of the Java VM. Java byte codes help make "write once, run anywhere" possible. You can compile your program into byte codes on any platform that has a Java compiler. The byte codes can then be run on any implementation of the Java VM. That means that as long as a computer has a Java VM, the same program written in the Java programming language can run on Windows 2000, a Solaris workstation, or on an iMac.



4.1.3 The Java Platform

A platform is the hardware or software environment in which a program runs. We've already mentioned some of the most popular platforms like Windows 2000, Linux, Solaris, and MacOS.

Most platforms can be described as a combination of the operating system and hardware. The Java platform differs from most other platforms in that it's a software- only platform that runs on top of other hardware-based platforms. The Java platform has two components: The Java Virtual Machine (Java VM) The Java Application Programming Interface (Java API) You've already been introduced to the Java VM. It's the base for the Java platform and is ported onto various hardware-based platforms. The Java API is a large collection of ready-made software components that provide many useful capabilities, such as graphical user interface (GUI) widgets. The Java API is grouped into libraries of related classes and interfaces; these libraries are known as packages. The next section, What Can Java Technology Do? Highlights what functionality some of the packages in the Java API provide. The following figure depicts a program that's running on the Java platform. As the figure shows, the Java API and the virtual machine insulate the program from the hardware.



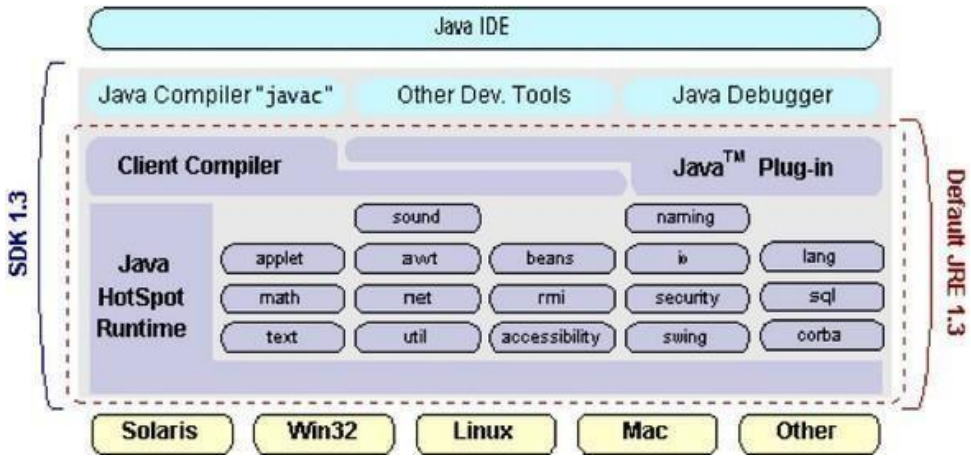
Native code is code that after you compile it, the compiled code runs on a specific hardware platform. As a platform-independent environment, the Java platform can be a bit slower than native code. However, smart compilers, well-tuned interpreters, and just- in-time byte code compilers can

bring performance close to that of native code without threatening portability. What Can Java Technology Do? The most common types of programs written in the Java programming language are applets and applications. If you've surfed the Web, you're probably already familiar with applets. An applet is a program that adheres to certain conventions that allow it to run within a Java-enabled browser. However, the Java programming language is not just for writing cute, entertaining applets for the Web. The general-purpose, high-level Java programming language is also a powerful software platform. Using the generous API, you can write many types of programs. An application is a standalone program that runs directly on the Java platform. A special kind of application known as a server serves and supports clients on a network. Examples of servers are Web servers, proxy servers, mail servers, and print servers. Another specialized program is a servlet. A servlet can almost be thought of as an applet that runs on the server side. Java Servlets are a popular choice for building interactive web applications, replacing the use of CGI scripts. Servlets are similar to applets in that they are runtime extensions of applications. Instead of working in browsers, though, servlets run within Java Web servers, configuring or tailoring the server.

How does the API support all these kinds of programs? It does so with packages of software components that provides a wide range of functionality. Every full implementation of the Java platform gives you the following features:

The essentials: Objects, strings, threads, numbers, input and output, data structures, system properties, date and time, and so on. Applets: The set of conventions used by applets. Networking: URLs, TCP (Transmission Control Protocol), UDP (User Data gram Protocol) sockets, and IP (Internet Protocol) addresses. Internationalization: Help for writing programs that can be localized for users worldwide. Programs can automatically adapt to specific locales and be displayed in the appropriate language. Security: Both low level and high level, including electronic signatures, public and private key management, access control, and certificates. Software components: Known as JavaBeans TM, can plug into existing component architectures. Object serialization: Allows lightweight persistence and communication via Remote Method Invocation (RMI). Java Database Connectivity (JDBCTM): Provides uniform access to a wide range of relational databases. The Java platform also has APIs for 2D and 3D

servers, colla boration, telephony, speech, animationg,rapnhdicsm, oraec. cTeshseibfiloitllyo,wing figure depicts what is included in the Java 2 SDK.



How Will Java Technology Change My Life

We can't promise you fame, fortune, or even a job if you learn the Java programming language. Still, it is likely to make your programs better and requires less effort than other languages. We believe that Java technology will help you do the following:

Get started quickly: Although the Java programming language is a powerful object-oriented it'easy to learn, especially for programmers already familiar with C or C++.

Write less code: Comparisons of program metrics (class counts, method counts, and so on) suggest that a program written in the Java programming language can be four times smaller than the same program in C++.

Write better code: The Java programming language encourages good coding practices, and its garbage collection helps you avoid memory leaks. Its object orientation, its JavaBeans component architecture, and its wide-ranging, easily extendible API let you reuse other people's tested code and introduce fewer bugs.

Develop programs more quickly: Your development time may be as much as twice as fast versus writing the same program in C++. Why? You write fewer lines of code and it is a simpler programming language than C++.

Avoid platform dependencies with 100% Pure Java: You can keep your program portable by avoiding the use of libraries written in other languages. The 100% Pure JavaTM Product Cpaeprteifrisc,ation Program

has a repository of historical process brocmhuarneusa, lasn, d wsihmitielar materials online. Write once, run

anywhere: Because 100% Pure Java programs are compiled into machine- independent byte codes, they run consistently on any Java platform. Distribute software more easily: You can upgrade applets easily from a central server. Applets take advantage of the feature of allowing new classes to be loaded “on the fly,”

without recompiling the entire program.

ODBC Microsoft Open Database Connectivity (ODBC) is a standard programming interface for application developers and database systems providers. Before ODBC became a de facto standard for Windows programs to interface with database systems, programmers had to use proprietary languages for each database they wanted to connect to. Now, ODBC has made the choice of the database system almost irrelevant from a coding perspective, which is as it should be. Application developers have much more important things to worry about than the syntax that is needed to port their program from one database to another when business needs suddenly change. Through the ODBC Administrator in Control Panel, you can specify the particular database that is associated with a data source that an ODBC application program is written to use. Think of an ODBC data source as a door with a name on it. Each door will lead you to a particular

database. For example, the data source named Sales Figures might be a SQL Server database, whereas the Accounts Payable data source could refer to an Access database. The physical database referred to by a data source can reside anywhere on the LAN.

The ODBC system files are not installed on your system by Windows 95. Rather, they are installed when you setup a separate database application, such as SQL Server Client or Visual Basic 4.0. When the ODBC icon is installed in Control Panel, it uses a file called ODBCINST.DLL. It is also possible to administer your ODBC data sources through a stand-alone program called ODBCADM.EXE. There is a 16-bit and a 32-bit version of this program and each maintains a separate list of ODBC data sources.

From a programming perspective, the beauty of ODBC is that the application can be written to use the same set of function calls to interface with any data source, regardless of the database vendor. The source code of the application doesn’t change whether it talks to Oracle or SQL Server. We only mention these two as an example. There are ODBC drivers available for several dozen popular database systems. Even Excel spreadsheets and plain text files can be turned into data

sources. The operating system uses the Registry information written by ODBC Administrator to determine which low-level ODBC drivers
are needed to talk to the data source (such as the interface to Oracle or SQL Server). The loading of the ODBC drivers is transparent to the ODBC application program. In a client/server environment, the ODBC API even handles many of the network issues for the application programmer.
The advantages of this scheme are so numerous that you are probably thinking there must be some catch. The only disadvantage of ODBC is that it isn't as efficient as talking
directly to the native database interface. ODBC has had many detractors make the charge that it is too
slow. Microsoft has always claimed that the critical factor in performance is the quality of the software that is used. In our humble opinion, this is true. The availability of good ODBC drivers has improved a great deal recently. And anyway, the criticism about performance is somewhat analogous to those who said that compilers would never match the speed of pure assembly language. Maybe not, but the compiler (or ODBC) gives you the opportunity to write cleaner programs, which means you finish compusteorosnger.t Mfaesatenrwehvileer,y year.
JDBC In an effort to set an independent database standard API for Java; Sun Microsystems developed Java Database Connectivity, or JDBC. JDBC offers a generic SQL database access mechanism that provides a consistent interface to a variety of RDBMSs. This consistent interface is achieved through the use of "plug-in" database connectivity modules, or drivers. If a database vendor wishes to have JDBC support, he or she must provide the driver for each platform that the database and Java run on.

To gain a wider acceptance of JDBC, Sun based JDBC's framework on ODBC. As you discovered earlier in this chapter, ODBC has widespread support on a variety of platforms. Basing JDBC on ODBC will allow vendors to bring JDBC drivers to market much faster than developing a completely new connectivity solution. JDBC was announced in March of 1996. It was released for a 90 day public review that ended June 8, 1996. Because of user input, the final JDBC v1.0 specification was released soon after. The remainder of this section will cover enough information about JDBC for you to know what it is about and how to use it effectively. This is by no means a complete overview of JDBC. That would fill an entire book.

JDBC **Goals** Fitesw software

packages are designed without goals in mind. JDBC is one that, because of many goals, drove the development of the API. These goals, in conjunction with early rbeuviiledwinegr feedback, have finalized the JDBC class library into a solid framework for database applications in Java.

Twhhey goals that were set for JDBC are important. They will give you some insight as to certain classes and functionalities behave the way they do. The eight design goals for
JDBC
are as follow

4.1.4. SQL Level API

The designers felt that their main goal was to define a SQL interface for Java. Although not the lowest database interface level possible, it is at a low enough level for higher-level tools and APIs to be created. Conversely, it is at a high enough level for application programmers to use it confidently. Attaining this goal allows for future tool vendors to "generate" JDBC code and to hide many of JDBC's complexities from the end user.

SQL Conformance:

SQL syntax varies as you move from database vendor to database vendor. In an effort to support a wide variety of vendors, JDBC will allow any query statement to be passed through it to the underlying database driver. This allows the connectivity module to handle non-standard functionality in a manner that is suitable for its users. JDBC must be implemental on top of common database interfaces The JDBC SQL API must "sit" on top of other common SQL level APIs. This goal allows JDBC to use existing ODBC level drivers by the use of a software interface. This interface would translate JDBC calls to ODBC and vice versa. Provide a Java interface that is consistent with the rest of the Java system Because of Java's acceptance in the user community thus far, the designers feel that they should not stray from the current design of the core Java system. This goal probably appears in all software design goal listings. JDBC is no exception. Sun felt that the design of JDBC should be very simple, allowing for only one method of completing a task per mechanism. Allowing duplicate functionality only serves to confuse the users of the API. Use strong, static typing wherever possible Strong typing allows for more error checking
to be done at compile time; also, less error appear at runtime. Keep the common cases simple
Because DmEoLrEeTEo'fsteandtUhPaDnAnTEot's, , ttheeseuqsuearlieSsQshLoculadllbse ussimedplebtyo ptheerfoprmrogrammer are simple SHEoLwEeCvTe'rs, ,mIoNrSeEcRoTm'sp,lewxitShQJLDsBtaCt.e ments should also be possible.

Finally we decided to proceed the implementation using Java Networking. And for dynamically updating the cache table we go for MS Access database.

Java ha two things: a programming language and a platform.

Java is a high-level programming language that is all of the following

Simple Architecture-neutral

Object-oriented Portable

Distributed High-performance

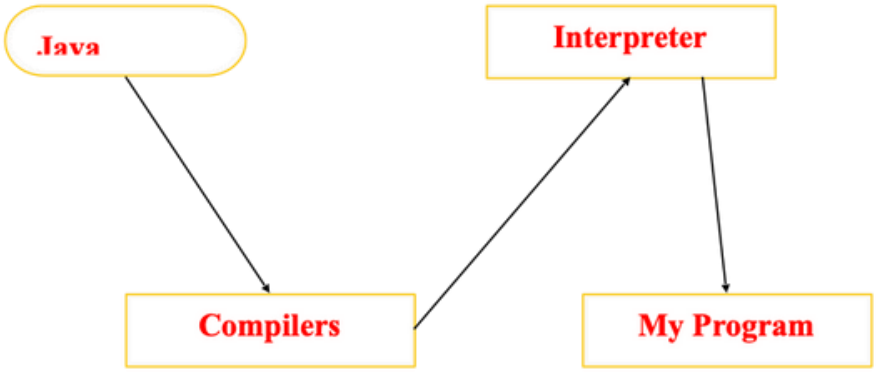
Interpreted multithreaded

Robust Dynamic

Secure

Java is also unusual in that each Java program is both compiled and interpreted. With a compile you translate a Java program into an intermediate language called Java byte codes the platform-independent code instruction is passed and run on the computer.

Compilation happens just once; interpretation occurs each time the program is executed. The figure illustrates how this works.



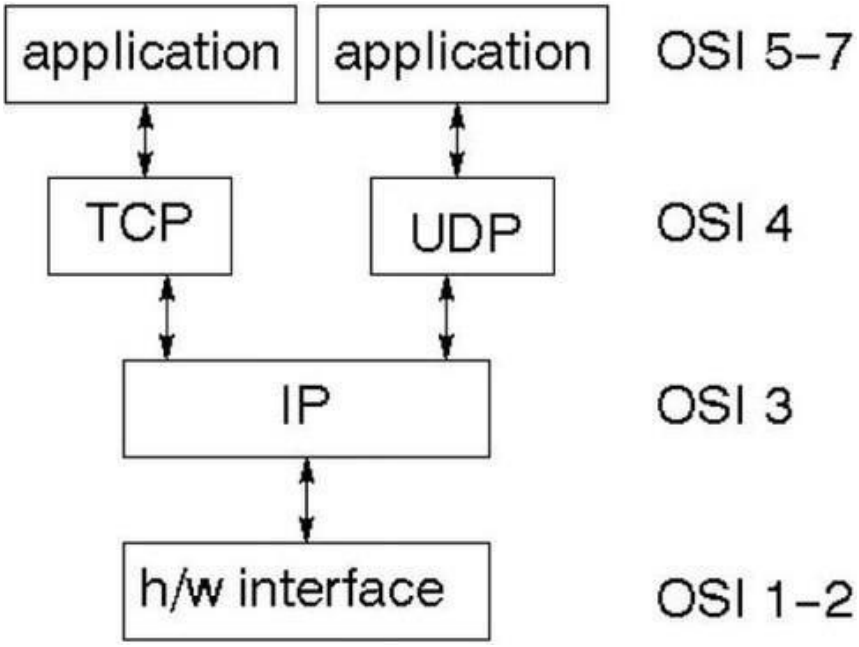
You can think of Java byte codes as the machine code instructions for the Java Virtual Machine (Java VM). Every Java interpreter, whether it's a Java development tool or a Web browser that can run Java applets, is an implementation of the Java VM. The Java VM can also be implemented in hardware. Java byte codes help make "write once, run anywhere" possible. You can compile your Java program into byte codes on my platform that has a Java compiler. The byte codes can then be

run any implementation of the Java VM. For example, the same Java program can run Windows NT, Solaris, and Macintosh.

4.2 Networking

TCP/IP stack

The TCP/IP stack is shorter than the OSI one:



TCP is a connection-oriented protocol;

UDP (User Datagram Protocol) is a connectionless protocol.

IP datagram's

The IP layer provides a connectionless and unreliable delivery system. It considers each datagram independently of the others. Any association between datagram must be supplied by the higher layers. The IP layer supplies a checksum that includes its own header. The header includes the source and destination addresses. The IP layer handles routing through an Internet. It is also responsible for breaking up large datagram into smaller ones for transmission and reassembling them at the other end.

UDP

UDP is also connectionless and unreliable. What it adds to IP is a checksum for the contents of the datagram and port numbers. These are used to give a client/server model - see later.

TCP

TCP supplies logic to give a reliable connection-oriented protocol above IP. It provides a virtual circuit that two processes can use to communicate.

Internet addresses

In order to use a service, you must be able to find it. The Internet uses an address scheme for machines so that they can be located. The address is a 32 bit integer which gives the IP address. This encodes a network ID and more addressing. The network ID falls into various classes according to the size of the network address. Network address

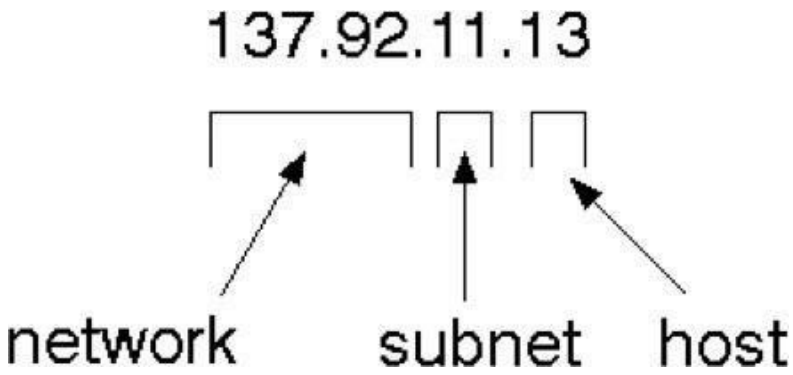
Class A uses 8 bits for the network address with 24 bits left over for other addressing. Class B uses 16 bit network addressing. Class C uses 24 bit network addressing and class D uses all 32.

Subnet address

Internally, the UNIX network is divided into sub networks. Building 11 is currently on one sub network and uses 10-bit addressing, allowing 1024 different hosts. **Host address**

8 bits are finally used for host addresses within our subnet. This places a limit of 256 machines that can be on the subnet.

4.3 Total address



Port addresses

A service exists on a host, and is identified by its port. This is a 16 bit number. To send a message to a server, you send it to the port for that service of the host that it is running on. This is not location transparency! Certain of these ports are "well known".

Sockets

A socket is a data structure maintained by the system to handle network connections. A socket is created using the call socket. It returns an integer that is like a file descriptor. In fact, under Windows, this handle can be used with Read File and Write File functions. #include <sys/types.h> #include <sys/socket.h>

int socket(int family, int type, int protocol);

Here "family" will be AF_INET for IP communications, protocol will be zero, and type will depend on whether TCP or UDP is used. Two processes wishing to communicate over a network create a socket each. These are similar to two ends of a pipe - but the actual pipe does not yet exist.

JFree Chart

JFreeChart is a free 100% Java chart library that makes it easy for developers to display professional quality charts in their applications. JFreeChart's extensive feature set includes:

A consistent and well-documented API, supporting a wide range of chart types;

A flexible design that is easy to extend, and targets both server-side and client-side applications;

Support for many output types, including Swing components, image files (including PNG and JPEG), and vector graphics file formats (including PDF, EPS and SVG);

JFreeChart is "open source" or, more specifically, free software. It is distributed under the terms of the GNU Lesser General Public Licence (LGPL), which permits use in proprietary applications.

1. Map Visualizations

Charts showing values that relate to geographical areas. Some examples include: (a) population density in each state of the United States, (b) income per capita for each country in Europe, (c) life expectancy in each country of the world. The tasks in this project include:

Sourcing freely redistributable vector outlines for the countries of the world, states/provinces in particular countries (USA in particular, but also other areas); Creating an appropriate dataset interface (plus default implementation), a rendered, and integrating this with the existing XYPlot class in JFreeChart;

Testing, documenting, testing some more, documenting some more.

2. Time Series Chart Interactivity

Implement a new (to JFreeChart) feature for interactive time series charts --- to display a separate control that shows a small version of ALL the time series data, with a sliding "view" rectangle that allows you to select the subset of the time series data to display in the main chart.

3. Dashboards

There is currently a lot of interest in dashboard displays. Create a flexible dashboard mechanism that supports a subset of JFreeChart chart types (dials, pies, thermometers, bars, and lines/time series) that can be delivered easily via both Java Web Start and an applet.

4. Property Editors

The property editor mechanism in JFreeChart only handles a small subset of the properties that can be set for charts. Extend (or reimplement) this mechanism to provide greater end- user control over the appearance of the charts.

4.4. Tomcat 6.0 web server

Tomcat is an open source web server developed by Apache Group. Apache Tomcat is the servlet container that is used in the official Reference Implementation for the Java Servlet and JavaServer Pages technologies. The Java Servlet and JavaServer Pages specifications are developed by Sun under the Java Community Process. Web Servers like Apache Tomcat support only web components while an application server supports web components as well as business components

(BEAs Weblogic, is one of the popular application server).To develop a web application with jsp/servlet install any web server like JRun, Tomcat etc to run your application.

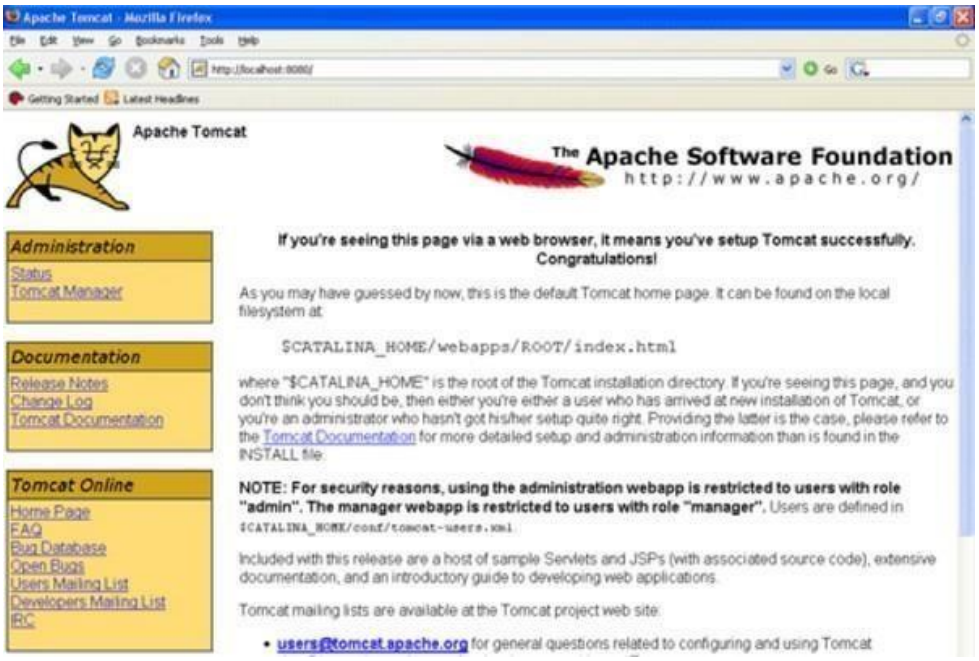


Fig Tomcat Webserver

4.5 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the

Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

4.6 TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised. Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify

Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot "see" into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach Field testing will be performed manually and functional tests will be written in detail.

Test objectives

All field entries must work properly.

Pages must be activated from the identified link. The entry screen, messages and responses must not be delayed.

Features to be tested

Verify that the entries are of the correct format No duplicate entries should be allowed All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects. The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

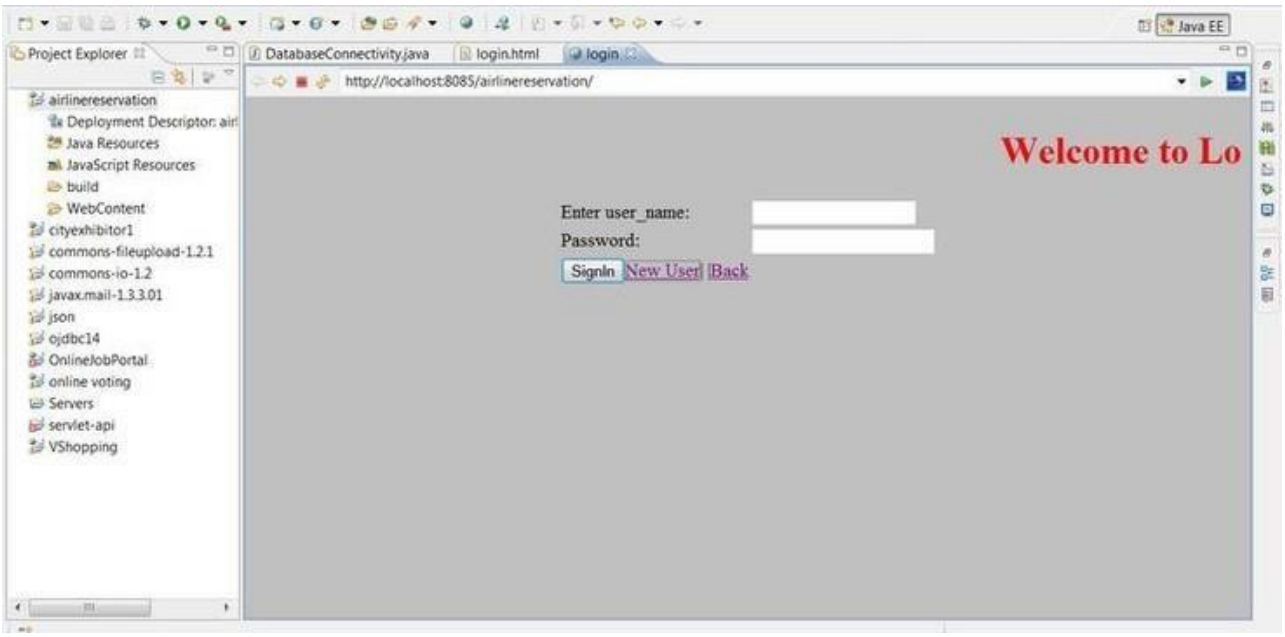
Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

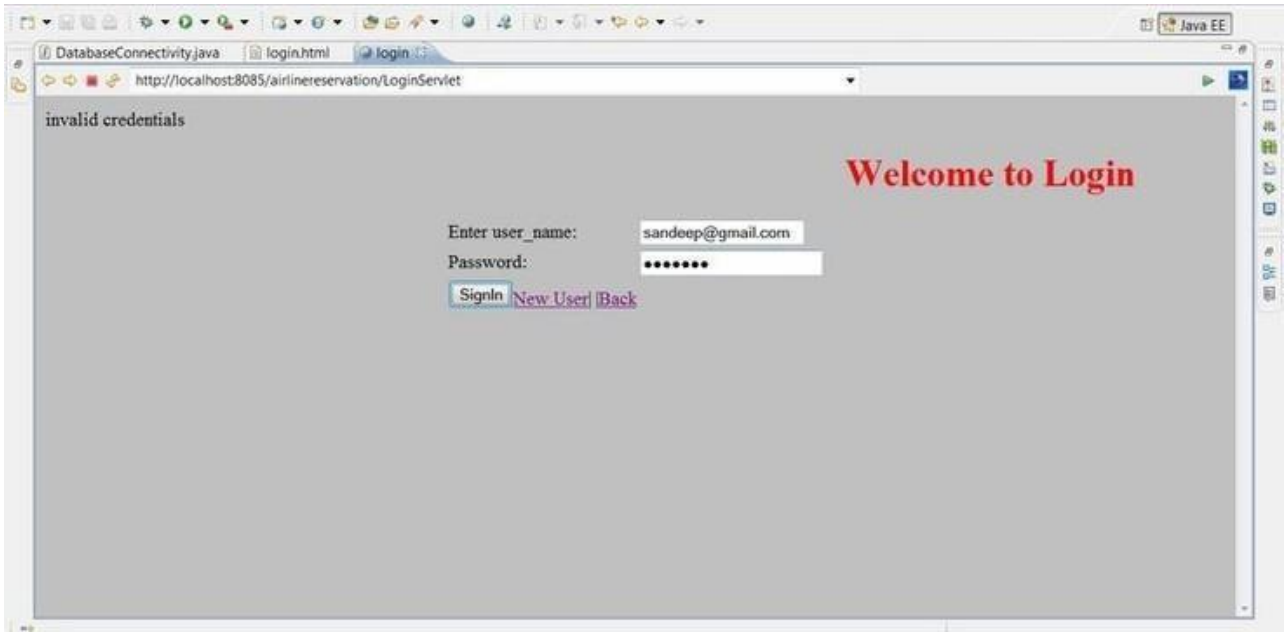
User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

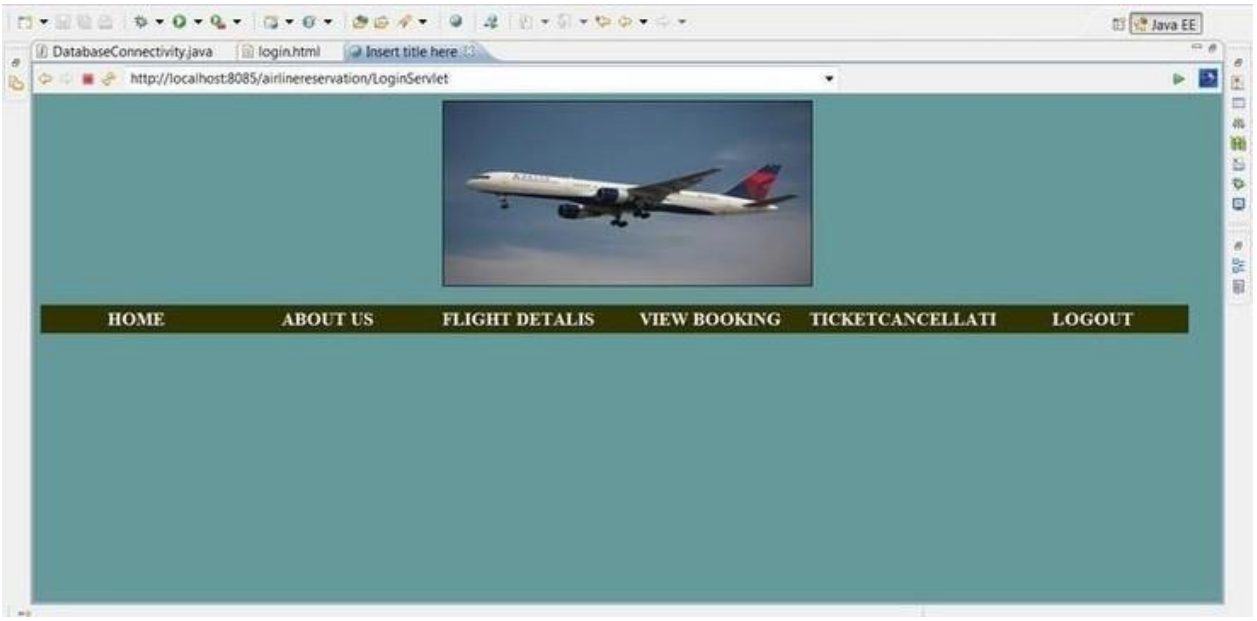
Test Results: All the test cases mentioned above passed successfully. No defects encountered

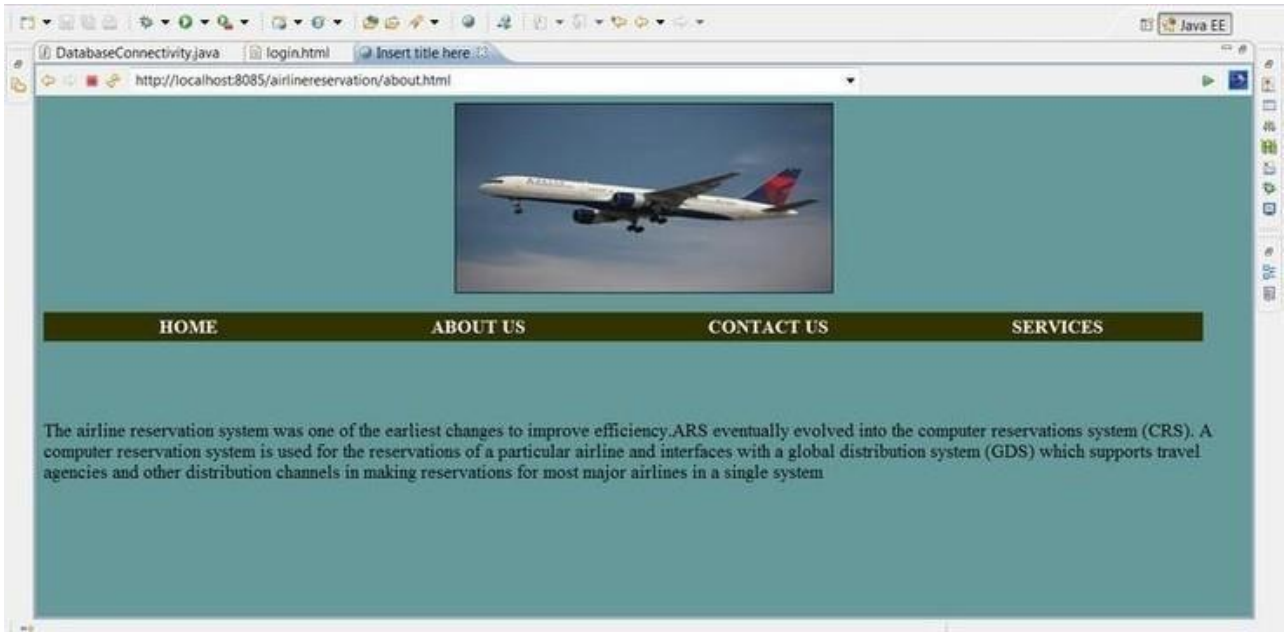
CHAPTER-5
OUTPUT&RESULT

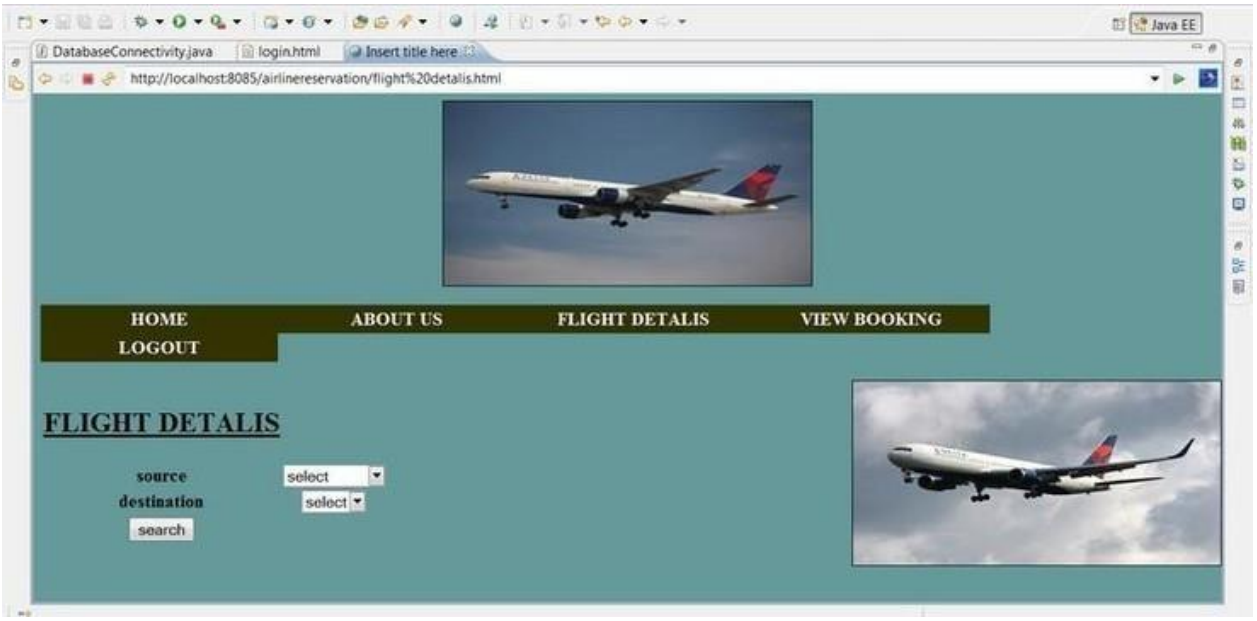


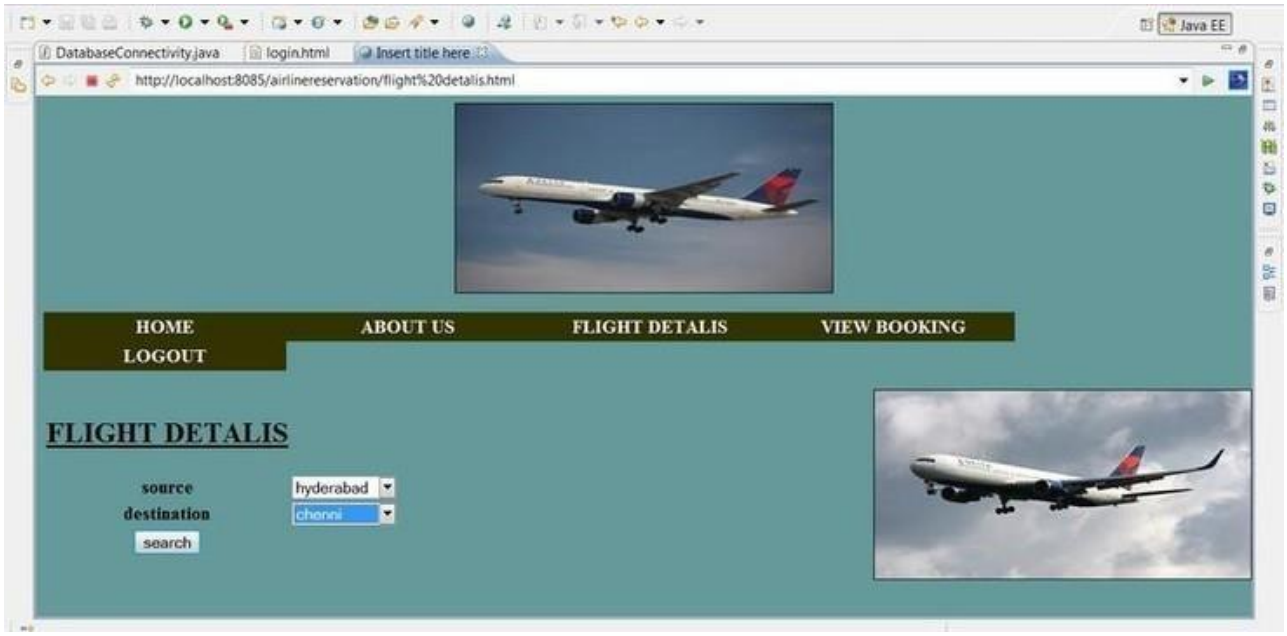



DatabaseConnectivity.java
login.html
login
http://localhost:8085/airlinereservation/LoginServlet
invalid credentials
Welcome to Login
Enter user_name:
sandeep@gmail.com
Password:
SignIn
New User
Back


DatabaseConnectivity.java
login.html
Insert title here
http://localhost:8085/airlinereservation/LoginServlet
HOME
ABOUT US
FLIGHT DETALIS
VIEW BOOKING
TICKETCANCELLATI
LOGOUT


DatabaseConnectivity.java
login.html
Insert title here
http://localhost:8085/airlinereservation/about.html
HOME
ABOUT US
CONTACT US
SERVICES
The airline reservation system was one of the earliest changes to improve efficiency.ARS eventually evolved into the computer reservations system (CRS). A computer reservation system is used for the reservations of a particular airline and interfaces with a global distribution system (GDS) which supports travel agencies and other distribution channels in making reservations for most major airlines in a single system


DatabaseConnectivity.java
login.html
Insert title here
http://localhost:8085/airlinereservation/flight%20detalis.html
HOME
ABOUT US
FLIGHT DETALIS
VIEW BOOKING
LOGOUT
FLIGHT DETALIS
source
select
destination
select
search



5.1. LIMITATIONS

Time and speed

These are dependent problems whereas by having slow speed that could be caused by having internet problem that could lead to delay of booking flight, check in leads to either delay flight or time wastage at the counter.

Security problem

Since the reservation occurs online there might be with security bleach or hacking of information thus leads to misleading of passengers.

Costly

By having to maintain data whereas each data from every branch has to be maintained and stored where as their copies have to be transferred to the concerned branch such as agencies, airports thus it becomes so expensive in maintenance and time wastage.

Skilled labour

Since the whole process requires people that either study airline course in case of maintenance of documents and operating it becomes a challenge to the ones that desire working to the sector with no high education experience.

5.2. FUTURE SCOPE OF THE PROJECT :

By having this system, it helps the airline companies to update and save in the flights and packages schedules at different timings whereas the customers get to obtain flight price, seat, meal and any other assistance that would help them have a comfortable flight. Moreover, for the future purpose this project is to reach and being understood even by the rural people whereas by development of this project we might use scan codes and applications, whereas all this is possible by having internet thus having sufficient infrastructure to provide them with internet that could help them save time moving to the city to meet the airline agencies.

5.3. CONCLUSION:

The airline reservation system was successfully designed and implemented with the intension to limit manual operations. This system is user friendly and very much flexible to non-programmers. It is also tested for maximum accuracy and quality validation. During this project designing, we have accomplished all the tasks and objectives to meet the organizational needs. The system will be used in searching, retrieving and generating processed information for the concerned request. With this proposed system are reduced errors due to human intervention and conveniently maintains any modifications to the flight and cancellations in the reservations made at any point of time. It not only provides flight details but also creates a platform to book tickets online, cancels or modifies ticket timings or dates and even informs about the number of people available on board and seats that are already booked.